

JEREMY ALLEN

AI Systems Architect | Automation & Platform Engineering

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PROFESSIONAL SUMMARY

Founder and hands-on AI systems builder focused on production automation, API integrations, agent-compatible commerce, cloud-edge deployment, and operator-facing software. Builds complete workflows from requirements and application logic through deployment, validation, checkout, and technical documentation. Recent work includes a live intent-driven commerce system connecting a branded Cloudflare Worker to Shopify's UCP catalog, cart, and checkout flow.

CORE TECHNICAL CAPABILITIES

AI automation & agent workflows • API integration • Python • JavaScript • HTML/CSS • Vite • Cloudflare Workers & Pages • Shopify commerce • UCP / MCP commerce flows • Git/GitHub • Linux / Pop!_OS / Ubuntu • Shell scripting • FastAPI • SQLite • Rust architecture • Go / terminal UI • Technical documentation & release engineering

PROFESSIONAL EXPERIENCE

Founder & AI Systems Architect | CrossingKey Intelligence | 2025–Present

- Architect and implement AI-enabled business systems, automation workflows, web applications, commerce infrastructure, and documented software releases for practical operator and small-business use.
- Build and deploy web and edge infrastructure using Cloudflare, Shopify, GitHub, Vite, JavaScript, Python, Linux tooling, and API-driven integrations.
- Translate business requirements into scoped technical systems, implementation plans, working interfaces, validation procedures, and customer-facing delivery paths.
- Design systems around inspectability, controlled automation, credential safety, reproducible releases, and clear handoffs between software, payment, delivery, and human review.

SELECTED PRODUCTION & ENGINEERING WORK

Commerce Systems Research, Remediation & Client-Safe Delivery | 2026

- Researched and remediated a live website/commerce stack, identifying stale storefront links, broken dynamic-search navigation, public configuration exposure, and disabled lead-capture paths.
- Produced a client-safe Shopify preview with 61 theme files and verified 26 routes, 0 Liquid errors, 0 missing pages, six policy links, and a passing desktop inspection before publication.
- Verified applied commerce corrections through read-back checks, including navigation, policy routing, page-body changes, unavailable collection cleanup, and stale-route elimination.
- Applied a zero-incremental-spend operating constraint by prioritizing existing, local, and open-source capabilities, checking billing risk, and separating verified work from actions requiring additional authorization.

CrossingKey Exchange | Agent-Compatible Commerce System | 2026

- Built and deployed a branded commerce interface at exchange.crossingkeyintelligence.com that accepts natural-language buyer intent and searches only the CrossingKey merchant catalog.
- Integrated Shopify UCP/MCP catalog discovery with real product variants, cart creation, checkout creation, and buyer handoff to Shopify-hosted payment.
- Deployed the Exchange as a Cloudflare Worker, mapped a custom production domain, exposed health/search/profile endpoints, and diagnosed a WAF managed-challenge conflict affecting API traffic.
- Integrated Exchange into the main CrossingKey Vite site as a first-class navigation and conversion destination, rebuilt the production site, and deployed it through Cloudflare Pages.

CrossingKey Web & Digital Commerce Platform

- Built and organized a branded web and product ecosystem using Shopify, Cloudflare, GitHub, JavaScript, Vite, HTML/CSS, and digital-product delivery tooling.

- Structured product discovery, checkout, policies, storefront navigation, deployment configuration, and buyer-facing commerce flows.

Agentic OS / Automation Architecture

- Designed a modular Rust-oriented agentic runtime architecture covering workflows, receipts, ledgers, plugins, QA, releases, and operator-visible execution.
- Developed reusable automation and documentation frameworks for controlled workflows, validation, rollback, release packaging, manifests, and checksums.

NavigatorFS / Terminal Control / Publication Tooling

- Developed Linux and Termux-oriented workspace tooling for diagnostics, validation, indexing, project inspection, and exports.
- Designed terminal control concepts in Go/Bubble Tea and publication systems for turning technical work into structured, verifiable deliverables.

CONTRACT FIT

Best suited for fixed-price implementation work where a client needs a working system delivered rather than advisory material alone: AI automation, internal tools, API-connected applications, agent workflows, SaaS/platform components, Shopify/commerce integrations, Cloudflare deployments, technical modernization, and production documentation.

PROFESSIONAL DEVELOPMENT

Self-directed AI systems and automation engineering, with continuous hands-on development across local Linux environments, cloud-edge deployment, commerce systems, agent workflows, and software release practices.